good philosophy

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MEMO TO: Mr. E. Bloch

SUBJECT: Revised Sigma Computer

Reference: First Draft: Development Program Proposal for a Revised Sigma Computer, Erich Bloch, April 7, 1959.

In the subject proposal, new components, new logical circuits, new packaging and a new systems design are proposed. The new technology, as presented, looks promising and should certainly be explored. Unfortunately, this technology does not solve the key bottleneck: memory. The new technology could be developed for an attachment to Sigma, similar to Harvest, or for a re-engineered machine module, for instance, the floating point unit. This development should avoid over-accelerated schedules; it should in no way interfere with development and production of 7030, Harvest, and 7000X. A revised computer, as proposed, would be too late and too slow. The reasons behind these conclusions are outlined below.

A. The Price of Being First

Right now Stretch is in the enviable position of constituting a real "IBM first". This achievement has not come free. First of all, it has taken vision, wisdom and great skill on the part of those who have directed the project. Secondly, as always, a price has been paid to achieve this goal. This inevitable price was mentioned in the subject proposal and can be summarized as follows:

1. Past Mistakes

Having learned by experience, we now wished we had chosen differently in the following areas:

a. Diode logic versus transistor logic.

b. PNP transistors only, versus PNP and NPN transistors.

c. Non-complementary logic versus complementary logic.

2. Surpassed Technology

The circuit technology for Stretch had to be selected at an early date. As the design proceeded, more attractive technology became commercially available, such as:

a. Mesa transistors.

b. Fast diodes.

IBM scored remarkably well with its memory technology. This success is very significant, since memory, not logic limits Stretch performance.

3. Patched Design

An advanced development invariably catches several groups off-guard. In order to proceed, compromise solutions must be accepted, such as:

a. Late and limited use of double cards.

b. Lack of adequate automated design.

c. Growing pains of standardization.

B. The Pride in Being First (and the dangers to which it leads)

Paying the price for being first does not imply that the benefits of being first will follow. The two procedures outlined below will quickly cause the benefits to be lost. They stem from the misconception that being first is all that counts.

1. Perpetual Postponement

This procedure is a classic in computer history. Illustrious machines, such as Babbage's Calculating Engine and the original Princeton machine, as well as obscure machines, such as the Raytheon computer, have been plagued by it. The procedure results from the desire to stay first in technology. One is unwilling to pay the price of surpassed technology. Rather, the technology is updated each time it is surpassed. The updating usually delays the schedule sufficiently to permit the technology to surpass the state of the art again. Thus, the process keeps repeating. Unwillingness to pay the full price of being first deprives one of all the benefits.

Now that computers are produced in series, the failure of reaching the production stage because of the development of revised models is equivalent to the case of perpetual postponement and leads to the same disastrous results.

2. Complacent Inaction

Too often a technical achievement is wasted because nothing is done with it (e.g., NORC). Machines stay "first" only for a short time. Mere inaction or delay in exploiting the advance results in lost opportunity and wasted effort.

Stretch has capitalized upon the shift in technology from tubes to transistors. A delayed project could take advantage only of an improvement in transistor technology. The latter advance is far less than the former, thus spoiling the competitive advantage.

C. The Prize for Being First

Persistent action is required to reap the profits from the successful fulfillment of the LASL contract.

Some profits are reaped already. The 2 u second memory proves very necessary to the 7090 in fighting its competition. However, in order to obtain the real profits from the LASL development (and the real answer to competition) other steps must be taken. The following three steps are recommended:

1. Establish the Product

The 7030 has probably a two-year lead over competition. This unrealized lead time should be frozen into a permanent advantage by selling, making and placing as many computers as possible. An example of the rewards from this type of vigorous follow-through is the 704, which became the standard in its performance class. This procedure exploits IBM's best assets: its sales force and its manufacturing facilities. It appeals to the customer in a very sensitive area: the desire for having the best and SHARE-ing common experience.

2. Widen the Product Range

In order to increase sales in the first decisive period, as wide a market as possible should be covered. A wide market is possible because:

a. The 7030 is a truly general-purpose computer.

b. A variable number of memory units can be attached.

c. The Harvest attachment allows an outward extension of the market.

d. The 7000X version allows a downward extension of the market. Customers who cannot afford a 7030 may buy this version now instead of Transac or 7090. When in 1963, say, a larger machine becomes desirable for such a customer, the step 7000X to 7030 is natural. When on the other hand a Transac or 7090 is bought now, the expansion to a larger machine will be towards the best machine of 1963, which no longer would be the 7030.

e. As yet unplanned units of different varieties, but attached in the Harvest manner can be provided.

3. Improve the Product

After the LASL computer is delivered and while the Bureau of Ships computer is finished, the 7000X version should be completed. Thus valuable design experience will be used to maximum advantage in the 7000X version. As the 7000X version is being completed, and while production is underway, the 7030 family (7030, - Harvest - 7000X) should be improved as a product. This improvement would result in smaller cost and size of the CPU. The following observations can be made about this process of product improvement as it relates to the proposal for a revised Sigma computer.

a. Modular improvement

The semi-modular construction of the CPU makes it possible to redesign selected parts of the CPU. These redesigned parts would be common to several or all members of the 7030 family. The 7030, Harvest system, and 7000X for instance, would all benefit from a redesigned Floating Point unit, I box, or VFL unit.

b. Schedule

The new circuits mentioned in the proposal (page 15) still require considerable evaluation (such as dc versus ac logic). Consequently, a relaxed time schedule is very desirable. Such a relaxed time schedule is possible only when production is not delayed by the introduction of the new circuits. When the new circuits are part of a product improvement program concurrent with production, the pressure to do a hasty job (do it late, do it fast, do it wrong, do it over) would be removed.

The production delay caused by a total CPU re-engineering effort would be considerable. In comparing the schedule proposed for the Revised Sigma with that for 7000X, it seems unrealistic that the same period of two years (for assembly and test completed) is proposed for both machines. The 7000X uses available circuits and requires only limited redesign; therefore it needs less time than the Revised Sigma proposal.

c. Manpower

As was pointed out in our recent conversation, the manpower requirements for 7000X and Revised Sigma are not in agreement.

	7000X 3/17/59	Revised Sigma 4/7/59
Component evaluation	None	2.00
Circuit design and packaging	None	17.75
Instruction redefinition	None	Not listed
Systems redesign	4.50	12.25
Assembly and test	21.50	11.50
Release	3.58	2.75
Total Manpower Requirements	29.58 *	46.25

* The 7000X estimates in addition list product engineering and manuals.

Extrapolation from the 7000X estimate would lead to a considerably increased estimate for the Revised Sigma. Early availability of automation of design may reduce the manual effort for both machines.

Re-engineering the CPU one unit at a time would reduce the manpower requirements. The relaxed time schedule and strict program compatibility would allow optimum use of automation of design, would omit an additional applied programming effort, and would reduce the customer engineering effort.

d. New Circuits

The proposed circuits look promising and should be explored. Unfortunately, they do not solve the memory bottleneck. Consequently, it is not possible to reduce look-ahead (16000 tr.) or omit one Y register (2150 tr.) as suggested. Omitting these units would reduce 7030 performance to one third of its present performance.

The new circuits would most clearly benefit the Floating Point multiplier. A redesigned floating point unit could perhaps omit the high-speed multiplier. Reduction of the adder from 96 to 48 bits almost certainly introduces incompatibility, which precludes this change.

In view of the fact that the 7030 right now falls short of Stretch objectives, at least part of the benefit of new circuits should be expressed in improved performance.

e. Instruction Redefinition

The 7000X report (12-31-'58, pages 18-20) states reasons for retaining strict program compatibility in the 7030 computer family. A possible reduction of 8900 transistors was shown to be insufficient to justify a serious loss in function and machine compatibility as well as an increase in charges for applied programming, development engineering, and customer engineering. The Revised Sigma proposal lists a subset of functions (except for Set and Wait) resulting in a component count reduction of 6700 transistors. The arguments of the 7000X report apply equally well to this subset and therefore need no repetition.

f. Hardware Reduction

The subject proposal shows that in each case n drift transistors can be replaced by 0.5 n diodes and 0.53 n mesa transistors. In addition, a hardware reduction might be obtained in the following areas:

Floating Point Unit	12000 tr.
I box	5000 tr.
Maintenance	2000 tr.
Total	19000 tr.

This reduction is 12% of the total transistor count. The reduction in the floating point unit alone is 31% and suggests this unit as the first object of selective product improvement.

g. Design Responsibility

A product improvement effort normally would be a manufacturing responsibility. The new circuit development, however, would have to be done by product development. If a re-engineering of selected machine sections does not provide sufficient funds and incentive, another attachment in the Harvest manner might prove to be a more suitable object.

D. Summary

New component, circuit and packaging techniques should be developed as a product improvement for the 7030 computer family. The product improvement should be done one unit at a time on a relaxed time schedule. Production of the 7030 family computers should proceed without any delay. A Revised Sigma Computer, as proposed, would have only one third of 7030 performance, would cause a serious production delay, and would add to product cost, thus spoiling our competitive advantage.

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